

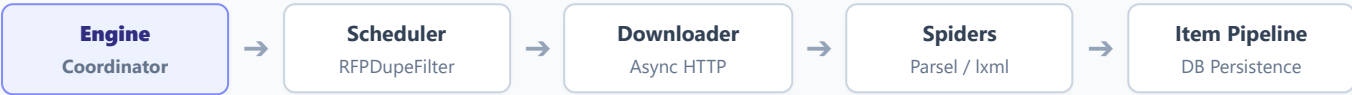
Scrapy // Asynchronous Crawling Framework

Reactive Reactor Topology, Parsing Algebra, AutoThrottle Engine & Hardened Settings

01. ARCHITECTURAL TOPOLOGY & REACTIVE DATA FLOW

Non-Blocking Twisted / Asyncio Finite State Machine

The Scrapy Engine coordinates non-blocking I/O event loops, passing Requests and Responses through middleware pipelines without blocking execution threads.



02. CONTROL MATRIX & CLI COMMANDS

Project-Scoped Execution

WORKSPACE

COMMAND	SYNTAX EXAMPLE	OPERATIONAL SEMANTICS
crawl	<code>scrapy crawl spider -O out.jsonl</code>	Boots reactor, runs spider, stream exports.
parse	<code>scrapy parse --spider=sp -c parse_item <url></code>	Unit tests spider callback routines directly.
genspider	<code>scrapy genspider -t crawl name domain.com</code>	Scaffolds template spider code.
check	<code>scrapy check <spider_name></code>	Runs contract unit tests defined in docstrings.

Global Diagnostics

STANDALONE

COMMAND	SYNTAX EXAMPLE	OPERATIONAL SEMANTICS
shell	<code>scrapy shell "<url>" --nolog</code>	Interactive IPython REPL for testing selectors.
fetch	<code>scrapy fetch --headers "<url>"</code>	Fetches raw HTTP headers & body via Downloader.
view	<code>scrapy view "<url>"</code>	Renders page in local browser to test JS reliance.
runspider	<code>scrapy runspider standalone_spider.py</code>	Executes single-file self-contained spider.

EXTRACTION OBJECTIVE	CSS3 SYNTAX	XPATH 1.0 EQUIVALENT	ALGORITHMIC NUANCE
Attribute Value	<code>a::attr(href)</code>	<code>//a/@href</code>	XPath supports <code>contains()</code> and <code>starts-with()</code> .
Text Extraction	<code>div.title::text</code>	<code>//div[@class='title']/text()</code>	XPath supports recursive string collation: <code>string(//div)</code> .
Ancestor Traversal	<i>Not supported</i>	<code>//input[@id='x']/ancestor::form</code>	Permits bidirectional graph traversal across DOM tree.
Regex Extraction	<code>css('h1::text').re(r'\d+')</code>	<code>xpath('//h1/text()').re_first(r'\d+')</code>	Direct node regex matching without Python string allocation.

03. AUTOTHROTTLE ENGINE & DIAGNOSTIC MATRIX

Adaptive AutoThrottle Algorithm

DYNAMIC BACKOFF

Adjusts download delay dynamically according to target server latency and response times.

1. Latency Calculation:

Measures round-trip time (RTT) for each HTTP response.

2. Concurrency Target:

Maintains `AUTOTHROTTLE_TARGET_CONCURRENCY`.

3. Proportional Delay:

`delay = latency / target_concurrency`.

4. Safety Bounds:

Clamped within `START_DELAY` and `MAX_DELAY`.

5. Backoff on Overload:

Auto-throttles up upon receiving 503 or 429 status codes.

Failure Remediation

DIAGNOSTICS

Cloudflare 403

Use residential proxies, modern Sec-CH headers, or Playwright sidcar.

Reactor Error

Call `asyncio reactor.install()` prior to importing reactor.

OOM / Memory Balloon

Set `DEPTH_PRIORITY = 1` and enforce `MEMUSAGE_LIMIT_MB`.

DNS Timeouts

Enable `DNSCACHE_ENABLED = True` and reduce `DOWNLOAD_TIMEOUT`.

04. HARDENED PRODUCTION CONFIGURATION (SETTINGS.PY)

```
# =====
# PhTex High-Performance Production Scrapy settings.py
# =====
TWISTED_REACTOR = "twisted.internet.asyncioreactor.AsyncioSelectorReactor"
CONCURRENT_REQUESTS = 32 # Max concurrent asynchronous I/O Loops
CONCURRENT_REQUESTS_PER_DOMAIN = 16
DOWNLOAD_TIMEOUT = 15 # Prevent hung socket resource starvation

# Adaptive AutoThrottle Rate Control
AUTOTHROTTLE_ENABLED = True
AUTOTHROTTLE_START_DELAY = 1.0
AUTOTHROTTLE_MAX_DELAY = 10.0
AUTOTHROTTLE_TARGET_CONCURRENCY = 2.0

# Retry Resiliency & Memory Protection
RETRY_ENABLED = True
RETRY_TIMES = 3
RETRY_HTTP_CODES = [500, 502, 503, 504, 522, 524, 408, 429]
MEMUSAGE_ENABLED = True
MEMUSAGE_LIMIT_MB = 2048 # Terminate cleanly before OS OOM killer

# DNS Resolution Caching & Deduplication
DNSCACHE_ENABLED = True
DNSCACHE_SIZE = 10000
REQUEST_FINGERPRINTER_IMPLEMENTATION = "2.7"
DUPEFILTER_CLASS = "scrapy.dupefilters.RFPDupeFilter"

DEFAULT_REQUEST_HEADERS = {
    "Accept": "text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8",
    "Accept-Language": "en-US,en;q=0.9",
    "Upgrade-Insecure-Requests": "1",
}
```